7.4 Adding and Subtracting Rational Expressions

Name _____ Period ____ Date _____

Algebra II 7.4 · Student sheet

I can…

- add and subtract rational expressions with like denominators.
- find the least common denominator and add or subtract with unlike denominators.
- simplify complex fractions.

Vocabulary · fill in the missing word

Least common denominator (LCD) — the least common _____ of the denominators

Complex fraction — a fraction with a fraction in its numerator, its _____, or both

Part 1 · Same move, new numbers

Work each pair side by side. Circle the step that is the same in both.

Numbers	Variables
$\frac{1}{6} + \frac{3}{8}$	$\frac{1}{6x} + \frac{3}{8x^2}$
$\frac{5}{12} - \frac{1}{18}$	$\frac{5}{12x} - \frac{1}{18x^2}$

Part 2 · Notes: fill in the blanks

Like denominators: add the _____; keep the _____.

When subtracting, put the second numerator in _____ and distribute the minus sign.

LCD: _____ each denominator and use each factor to its _____ power.

Complex fraction: multiply the top and bottom by the _____ of every small fraction.

Part 3 · Examples (work them with the class)

Example 1

Find the difference: $\frac{5}{x+2} - \frac{x+3}{x+2}$

Example 2

Find the least common multiple of $4x^2 - 16$ and $6x^2 - 24x + 24$.

Example 3

Find the sum: $\frac{7}{9x^2} + \frac{x}{3x^2 + 3x}$

Example 4

Find the difference: $\frac{x}{x^2 - x - 12} - \frac{2}{x + 3}$

Example 5

Simplify: $\left(\frac{5}{x + 4}\right) \div \left(\frac{1}{x + 4} + \frac{2}{x}\right)$

Part 4 · Partner practice

Partner 1

Find the sum: $\frac{2}{3x} + \frac{1}{6x^2}$

Partner 2

Simplify: $(\frac{1}{x} + \frac{1}{2}) \div (\frac{3}{x})$